

View Review

Paper ID

332

Paper Title

AI Driven Speech-Based Asthma Detection using Feature Optimization for Early Diagnosis

REVIEW QUESTIONS

1. Relevance of the Work with the theme of Conference ?

Yes

2. Novelty in the Paper

Yes

3. Comments to authors

☐ Strengths:

o The paper proposes a novel real-time asthma detection model using speech signals, leveraging the COSWARA dataset, which is highly relevant for non- invasive respiratory health monitoring.

o The integration of the Improved Social Spider Optimization with Genetic Algorithm (ISSO-G) for feature extraction and optimization is a significant contribution, enhancing discriminative feature selection.

o The use of a CNN for classification is well-justified, and the model achieves impressive performance (94.073% accuracy, up to 13.2% improvement over BiLSTMs), as shown in Table II and Fig. 3.

o The methodology is clearly structured, with detailed descriptions of data collection, feature extraction, optimization, and classification processes.

☐ Weaknesses:

o The introduction could better emphasize the specific challenges of speech- based asthma detection (e.g., noise robustness, demographic variability) to clarify the problem's significance.

o The dataset description in Table I is informative but lacks details on the number of asthmatics vs. non-asthmatic samples, which is critical for assessing class balance and model generalizability.

o The literature review is comprehensive but does not clearly articulate how the proposed ISSO-G+CNN model advances beyond cited works (e.g., [6], [8], [9]) in terms of specific algorithmic improvements.

o The mathematical formulations (Eq. 1–5) are complex and poorly explained, making it difficult to understand their role in the CNN architecture without additional context.

o Minor editorial issues, such as incomplete reference entries (e.g., [16]–[18]) and unclear figure captions (e.g., Fig. 1, Fig. 2), reduce the paper's polish.

☐ Suggestions for Improvement:

o Revise the introduction to clearly outline the unique challenges of speech- based